Transformations with Multiple Parameters

Precalculus   Functions and Composition

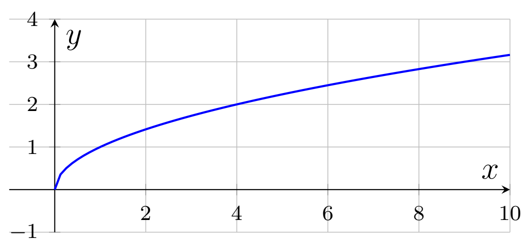
Transformations from a table, parameters inside an equation, and the order of composition

Directions: Every problem here has *more than one* parameter acting at once. Name each parameter before you compute, and keep the inside-the-function moves separate from the outside-the-function moves.

The parameter map — read every problem against this

$$g(x) = \underbrace{a}_{\substack{\text{vertical stretch} \\ \text{reflect if } a<0}} \cdot f\Big(\underbrace{b}_{\substack{\text{horizontal} \\ \text{compress by } 1/b}}(x - \underbrace{h}_{\substack{\text{shift right} \\ h \text{ units}}})\Big) + \underbrace{k}_{\substack{\text{shift up} \\ k \text{ units}}}$$

Outside the function (a, k) acts on the *output* and does what it says. **Inside** (b, h) acts on the *input* and does the opposite of what it looks like. The inside must be *factored* into the form $b(x-h)$ before h can be read.



The parent function $f(x) = \sqrt{x}$.

1. Let $f(x) = x^2$ and $g(x) = 3f(x-2) - 4$. Fill in the chain, then state $g(5)$.

x	$x-2$	$f(x-2)$	$3f(x-2)$	$3f(x-2)-4$
5				

Answer: ____________

2. Let $f(x) = x^2$ and $h(x) = -f(x+2) + 5$.
Which two parameters are non-trivial here, and what does each one do? Then find $h(1)$.

Answer: ________________

3. Let $g(x) = 2\sqrt{x-5} + 3$.
Solve $g(x) = 11$. Undo the outside parameters first, then the inside one, and say which step each parameter is undone in.

Answer: ________________

4. Let $y = 3\sin\left(2\left(x - \frac{\pi}{4}\right)\right) + 2$.
Identify the amplitude, the period, the phase shift and the midline. Then find y when $x = \dfrac{\pi}{2}$.

Answer: ________________

5. Two transformation maps: $s(x) = x + 5$ (a shift) and $d(x) = 3x$ (a stretch).
Write a simplified formula for $(d \circ s)(x) = d(s(x))$, and state in words which move happens first.

Answer: ________________

6. Using the same $s(x) = x + 5$ and $d(x) = 3x$:
For what value of x does $(d \circ s)(x) = 27$? Then explain in one sentence why $(s \circ d)(x)$ would *not* give the same answer.

Answer: ________________

7. Rewrite $y = 4\sin\left(3x - \frac{\pi}{2}\right)$ in the form $A\sin\big(B(x - C)\big)$.
State the phase shift C, and explain why it is not $\frac{\pi}{2}$.

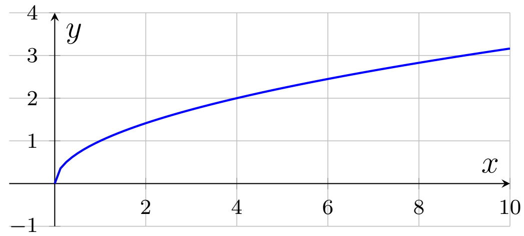
Answer: ________________

8. Let $f(x) = \sqrt{x}$ (graphed above) and $g(x) = -2f\big(4(x-1)\big) + 7$.
List all four parameters and what each one does, then compute $g(5)$.

Answer: ________________

9. Let $g(x) = 3\sqrt{x+6} - 9$.
Find the x-intercept of g. (Set the output to zero and undo the parameters in the reverse of the order that builds them.)

Answer: ________________



The parent function $f(x) = \sqrt{x}$.

10. Using the parent $f(x) = \sqrt{x}$ graphed above, sketch $y = -3f\left(\frac{x}{3} + 1\right) + 2$ on the grid below.

Before you sketch, factor the inside expression and list the four transformations in an order that produces the correct graph. Then explain why the horizontal shift is **3 units left** even though the inside reads "+1". Label the image of the parent's starting point on your sketch.